

MTH2021 Mathematical Methods 2

Self-study sheet 4 – solutions

1.

(a) Consider for definiteness the (easier) case of a simply connected region R . Suppose $f(z)$ is continuous in R and integrates to zero around every closed curve in R . First we show that $f(z)$ must possess an anti-derivative in R . Letting $f(z) = u(x, y) + iv(x, y)$:

$$\oint_C f(z) dz = \oint_C (u + iv)(dx + idy) \quad (1)$$

$$= \oint_C (udx - vdy) + i \int_C (vdx + udy) = 0 \quad (2)$$

for every closed curve C in R . Then the vector fields $(u, -v)$ and (v, u) are each an exact gradient:

$$u = \phi_x \quad , \quad -v = \phi_y \quad (3)$$

$$v = \psi_x \quad , \quad u = \psi_y \quad (4)$$

for suitable scalar fields ϕ and ψ . (Subscripts denote partial derivatives.) Hence

$$u = \phi_x = \psi_y \quad (5)$$

$$v = \psi_x = -\phi_y \quad (6)$$

But these are the Cauchy-Riemann equations. Hence

$$F(z) = \phi(x, y) + i\psi(x, y) \quad (7)$$

is analytic in R . Furthermore

$$\frac{dF}{dz} = \phi_x + i\psi_x \quad (8)$$

$$= u + iv = f \quad (9)$$

We know independently that analytic functions are infinitely differentiable. Hence f - the derivative of F - is analytic in R .

(b) Suppose $f(z)$ is analytic in a simply connected region R . Then its integrals around closed curves in R vanish and, by the above result, it possesses an anti-derivative $F(z) = \phi(x, y) + i\psi(x, y)$ in R with

$$f(z) = u(x, y) + v(x, y) = \frac{dF(z)}{dz} \quad (10)$$

Now consider the line integral of $f(z)$ between points a and b along a curve C in R :

$$\int_C f(z) dz = \int_C (udx - vdy) + i \int_C (vdx + udy) \quad (11)$$

$$= \int_C (\phi_x dx + \phi_y dy) + i \int_C (\psi_x dx + \psi_y dy) \quad (12)$$

$$= (\phi + i\psi) \Big|_a^b = F(b) - F(a) \quad (13)$$

2.

Let $f(z)$ be analytic within a circle of radius R centered at z_0 . Let z be inside this circle. Choose a circle C centered at z_0 with a radius r such that $|z - z_0| < r < R$. By Cauchy's integral formula

$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{w - z} dw \quad (14)$$

Now write

$$\begin{aligned} \frac{1}{w - z} &= \frac{1}{(w - z_0) - (z - z_0)} \\ &= \frac{1}{w - z_0} \frac{1}{1 - \frac{z - z_0}{w - z_0}} \end{aligned} \quad (15)$$

Since w in equation (14) runs around C , we have $|w - z_0| = r > |z - z_0|$, i.e.

$$\left| \frac{z - z_0}{w - z_0} \right| < 1 \quad (16)$$

We can then expand (15) binomially:

$$\frac{1}{w - z} = \frac{1}{w - z_0} \left(1 + \frac{z - z_0}{w - z_0} + \dots + \left(\frac{z - z_0}{w - z_0} \right)^n \right) + R_n \quad (17)$$

where the remainder R_n is

$$R_n = \frac{1}{w - z} \left(\frac{z - z_0}{w - z_0} \right)^{n+1} \quad (18)$$

Substituting into (14):

$$f(z) = \sum_{k=0}^n \frac{1}{2\pi i} (z - z_0)^k \oint_C \frac{f(w)}{(w - z_0)^{k+1}} dw + \tilde{R}_n \quad (19)$$

where

$$\tilde{R}_n = \frac{1}{2\pi i} \oint_C \frac{f(w)}{w - z} \left(\frac{z - z_0}{w - z_0} \right)^{n+1} dw \quad (20)$$

We aim to show that $|\tilde{R}_n| \rightarrow 0$ as $n \rightarrow \infty$:

$$|\tilde{R}_n| = \left| \frac{1}{2\pi i} \oint_C \frac{f(w)}{w - z} \left(\frac{z - z_0}{w - z_0} \right)^{n+1} dw \right| \quad (21)$$

$$\leq \frac{1}{2\pi} \oint_C \left| \frac{f(w)}{w - z} \right| \left| \frac{z - z_0}{w - z_0} \right|^{n+1} |dw| \quad (22)$$

which, in turn, dies out with increasing n , because $\left| \frac{z - z_0}{w - z_0} \right| < 1$. Hence the remainder disappears, and we are left with the series

$$f(z) = \sum_{k=0}^{\infty} (z - z_0)^k \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w - z_0)^{k+1}} dw \quad (23)$$

$$= \sum_{k=0}^{\infty} \frac{f^{(k)}(z_0)}{k!} (z - z_0)^k \quad (24)$$

For the final step above, we have used the Cauchy integral formula for the k -th derivative. This series converges for all z such that $|z - z_0| < R$ (because we can apply the above process to all such z). This is the Taylor series for a function analytic in a disc.

3.

(a) The integrand $\frac{\cos z}{(z+i)(z-2)}$ is analytic everywhere on and within C . Hence by Cauchy's theorem the integral is zero.

(b) Write the integral as

$$I = \oint_C \frac{\cos z}{(z - (-i))(z - 2)} dz = \oint_C \frac{f(z)}{z - z_0} dz \quad (25)$$

where

$$f(z) = \frac{\cos z}{z - 2} \quad \text{and} \quad z_0 = -i \quad (26)$$

The point $z = 2$ is outside C , while $z_0 = -i$ is inside C . Hence $f(z)$ is analytic everywhere on and within C , and by the Cauchy integral formula

$$I = 2\pi i f(z_0) = 2\pi i \frac{\cos(-i)}{-i - 2} = -2\pi i \frac{\cosh 1}{2 + i} \quad (27)$$

(c) Now $z = 2$ is inside C , while $z = -i$ is outside. Thus we write

$$I = \oint_C \frac{\cos z}{(z+i)(z-2)} dz = \oint_C \frac{f(z)}{z - z_0} dz \quad (28)$$

where

$$f(z) = \frac{\cos z}{z + i} \quad \text{and} \quad z_0 = 2 \quad (29)$$

$f(z)$ is analytic everywhere on and within C , and by the Cauchy integral formula

$$I = 2\pi i f(z_0) = 2\pi i \frac{\cos 2}{2 + i} \quad (30)$$

4.

(a) We use the Cauchy integral formula for the n -th derivative

$$\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0) \quad (31)$$

with $f(z) = \cos z$, $z_0 = 1$ and $n = 1$. We have $f^{(1)}(z) = f'(z) = -\sin z$ and so

$$\oint_C \frac{\cos z}{(z - 1)^2} dz = \frac{2\pi i}{1!} f'(1) = -2\pi i \sin 1 \quad (32)$$

(b) The integrand $\frac{\cos z}{(z - 1)^2}$ is analytic everywhere on and within C . Hence by Cauchy's theorem the integral is zero.

5.

(a) C is the circle of radius 2 centered at the origin. The integrand is analytic on and within C , and the integral is zero by Cauchy's theorem.

(b) We use the Cauchy integral formula for the n -th derivative

$$\oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz = \frac{2\pi i}{n!} f^{(n)}(z_0) \quad (33)$$

with $n = 100$ and $z_0 = 1$. For $f(z) = \exp(2z)$ the n -th derivative is $f^{(n)}(z) = 2^n \exp(2z)$ and thus

$$\oint_C \frac{\exp(2z)}{(z - 1)^{101}} dz = \frac{2\pi i}{100!} 2^{100} \exp 2 \quad (34)$$

(c) The integrand has a pole at $z = 4$ but that is outside C . Within and on C the integrand is analytic, and the integral is zero by Cauchy's theorem.

6.

If $f(z)$ is analytic on and within the contour C , and z_0 is inside C , then we have the Cauchy integral formula for the n -th derivative of $f(z)$ at $z = z_0$:

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z - z_0)^{n+1}} dz \quad (35)$$

If C encloses the origin and we choose $z_0 = 0$, then

$$f^{(n)}(0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{z^{n+1}} dz \quad (36)$$

If $f(z) = \exp(-irz)$, then $f^{(n)}(z) = (d/dz)^n \exp(-irz) = (-i)^n r^n \exp(-irz)$. At $z = 0$ this becomes $f^{(n)}(0) = (-i)^n r^n$. Hence

$$(-i)^n r^n = \frac{n!}{2\pi i} \oint_C \frac{\exp(-irz)}{z^{n+1}} dz \quad (37)$$

Now choose C to be the circle $|z| = 1$. For z on C , change variable to $z = \exp(i\theta) = \cos \theta + i \sin \theta$, $dz = i \exp(i\theta) d\theta$, $0 \leq \theta \leq 2\pi$. Substituting in:

$$\int_0^{2\pi} \exp(-ir(\cos \theta + i \sin \theta)) \frac{i \exp(i\theta)}{\exp(i(n+1)\theta)} d\theta = \frac{2\pi i}{n!} (-i)^n r^n \quad (38)$$

Rearranging:

$$\int_0^{2\pi} \exp(r \sin \theta) \exp(-i(n\theta + r \cos \theta)) d\theta = \frac{2\pi r^n}{n!} (-i)^n \quad (39)$$

Writing $(-i)^n = \exp(-in\pi/2)$ and taking real and imaginary parts of the two sides above:

$$\int_0^{2\pi} \exp(r \sin \theta) \cos(n\theta + r \cos \theta) d\theta = \frac{2\pi r^n}{n!} \cos(n\pi/2) \quad (40)$$

$$\int_0^{2\pi} \exp(r \sin \theta) \sin(n\theta + r \cos \theta) d\theta = \frac{2\pi r^n}{n!} \sin(n\pi/2) \quad (41)$$

7. Put $z - 1 = w$:

$$f(z) = \frac{\sin w}{1 + w} \quad (42)$$

We can now use the Taylor series for $\sin w$ about $w = 0$ and the binomial series for $1/(1 + w)$, which is absolutely convergent for $|w| < 1$ (i.e. $|z - 1| < 1$):

$$f(z) = \frac{\sin w}{1 + w} = (w - w^3/3! + w^5/5! \dots)(1 - w + w^2 - w^3 \dots) \quad (43)$$

$$= w - w^2 + (1 - 1/6)w^3 + (1/6 - 1)w^4 \dots \quad (44)$$

8.

(a) Let $w = z - 1$:

$$f(z) = \frac{1}{(z + 1)^2(z - 1)^2} \quad (45)$$

$$= \frac{1}{w^2(w + 2)^2} \quad (46)$$

$$= \frac{1}{4w^2(1 + w/2)^2} \quad (47)$$

$$= \frac{1}{4w^2} (1 - 2(w/2) + 3(w/2)^2 - 4(w/2)^3 \dots) \quad (48)$$

$$= \frac{1}{4(z - 1)^2} - \frac{1}{4(z - 1)} + \frac{3}{16} - \frac{z - 1}{8} \dots \quad (49)$$

The binomial expansion for $(1 + w/2)^{-2}$ above is valid for $|w/2| < 1$, or $|z - 1| < 2$, which is fulfilled in the region we are considering.

(b) Let $w = 1/z$:

$$f(z) = \frac{1}{(z + 1)^2(z - 1)^2} = \frac{1}{(z^2 - 1)^2} \quad (50)$$

$$= \frac{w^4}{(1 - w^2)^2} \quad (51)$$

$$= w^4(1 + 2w^2 + 3w^4 \dots) \quad (52)$$

$$= w^4 + 2w^6 + 3w^8 \dots \quad (53)$$

$$= \frac{1}{z^4} + \frac{2}{z^6} + \frac{3}{z^8} \dots \quad (54)$$

which is valid provided $|w^2| < 1$, or $|z| > 1$, as required.